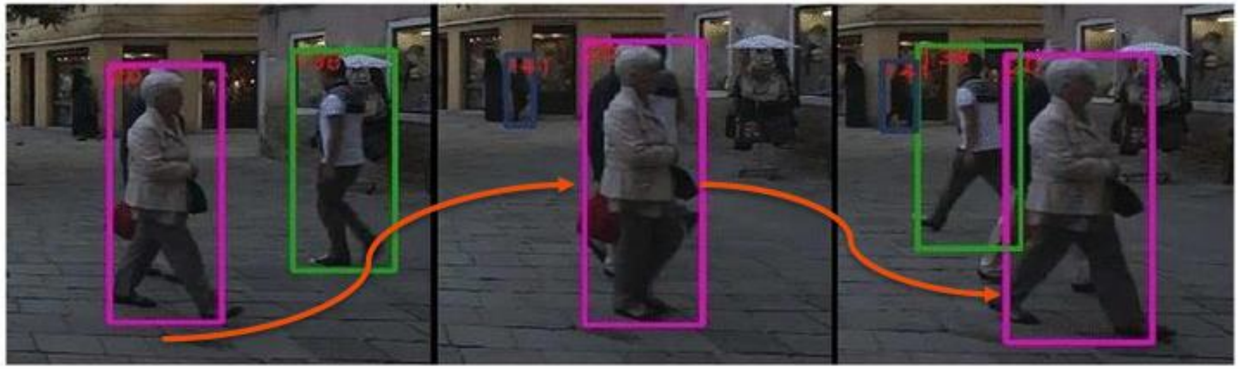
4 - Video Processing

Object Tracking, Temporal Modeling, Activity Recognition

Object Tracking

- Identify which parts of a video depict the same object in successive frames.

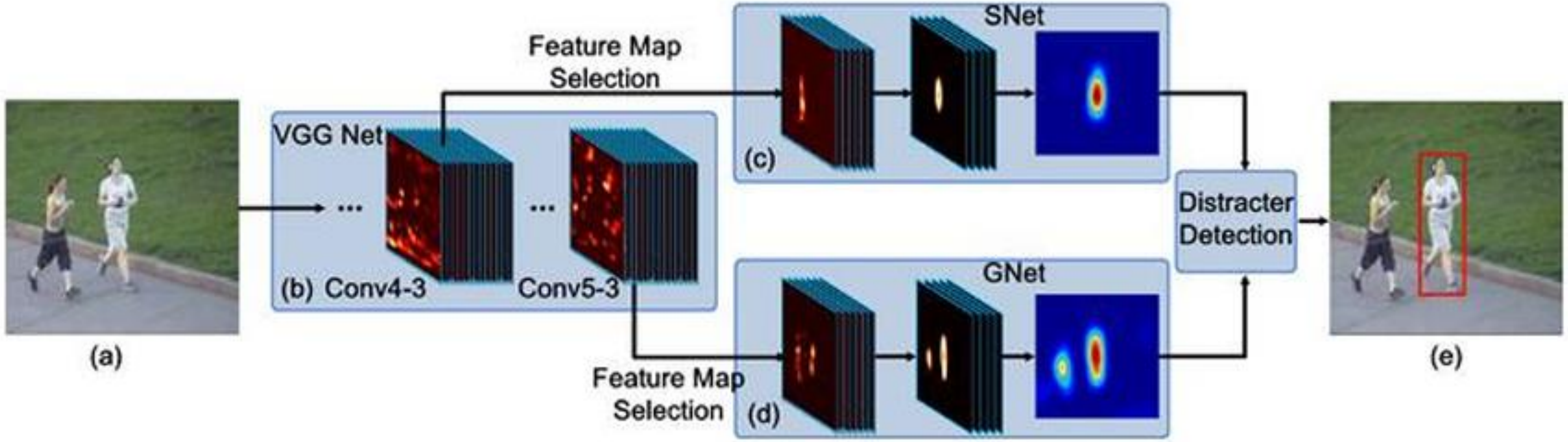


Frame i Frame i+1 Frame i+2

- Challenges include re-identification, occlusion, identity switching, motion blur, scale change, illumination, and the need for real-time tracking.

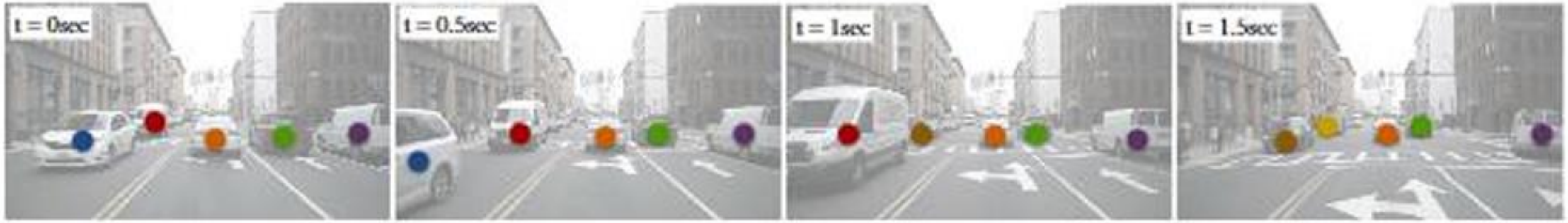
Tracking-by-Detection

- **Fully Convolution Network-based Tracker (FCNT)**: select discriminative feature maps and discard noisy ones for tracking.
- Built on top of a pre-trained VGG model.
- Two heat maps for *distracter prevention*: prevent the tracker from drifting to the background.



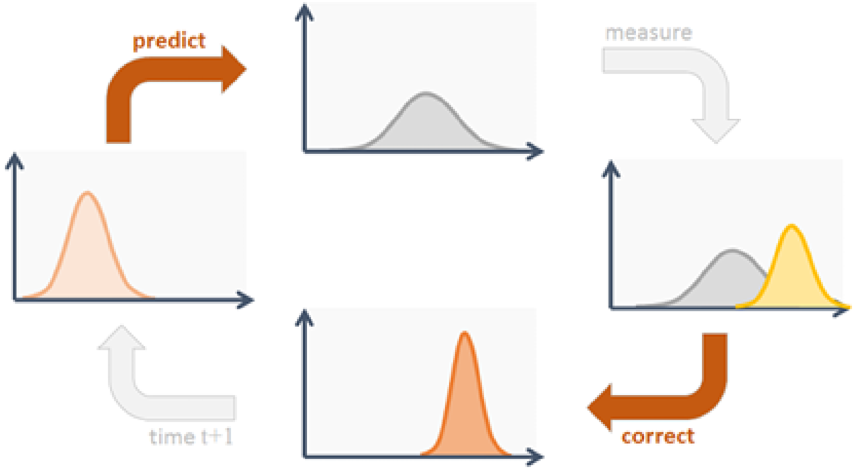
Tracking-by-Detection

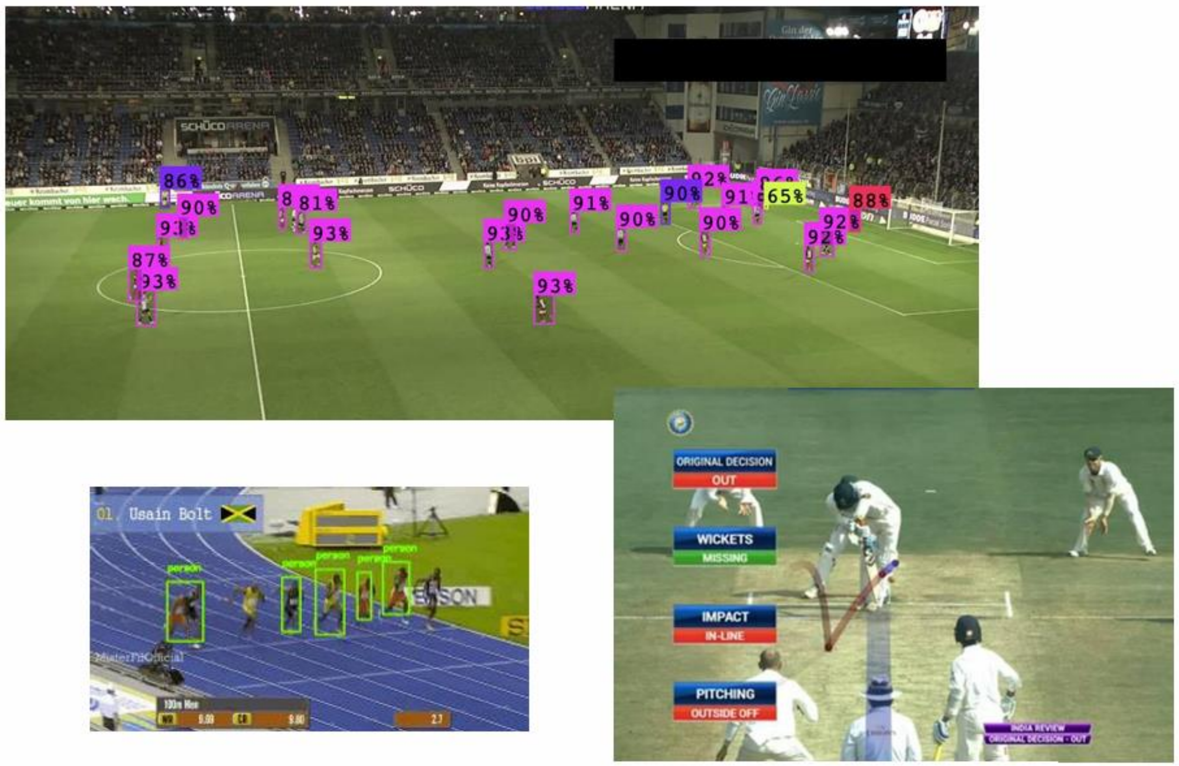
- **CenterTrack**: end-to-end simultaneous object detection and tracking.
- Each object is represented as a *point* at the center of its bounding box.
- Takes in 2 frames and a prior heatmap, produces detection and tracking offsets for the current frame in real time (22fps).



Tracking-by-Detection

- **DeepSORT**: real-time object detection using YOLO, motion estimation using a Kalman filter.

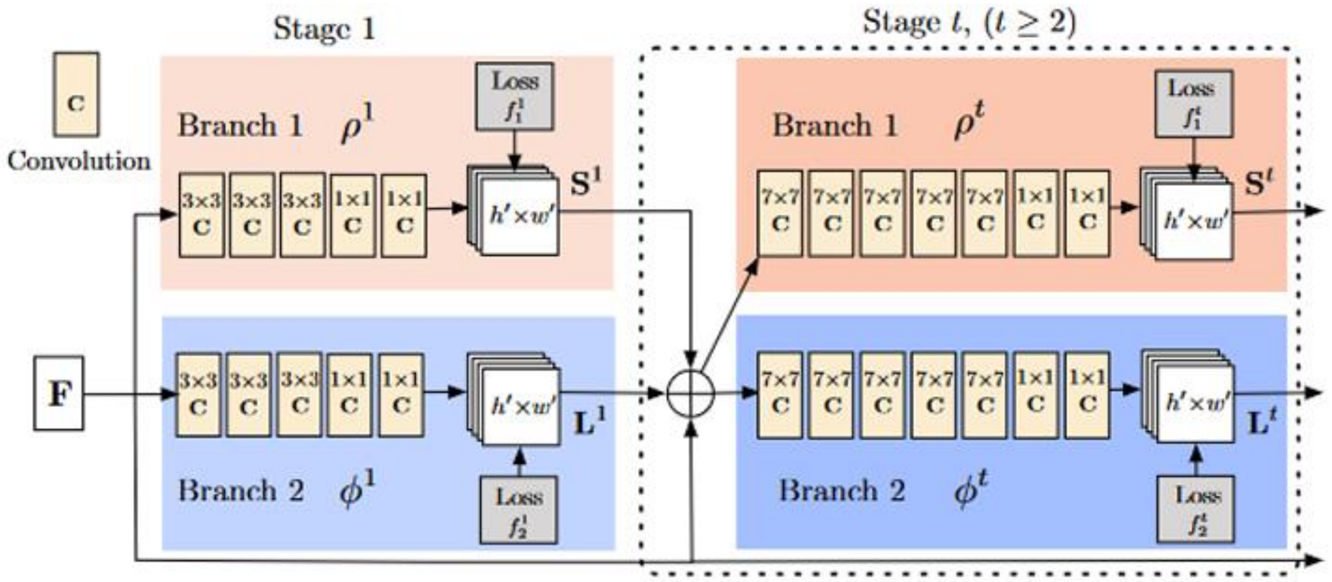


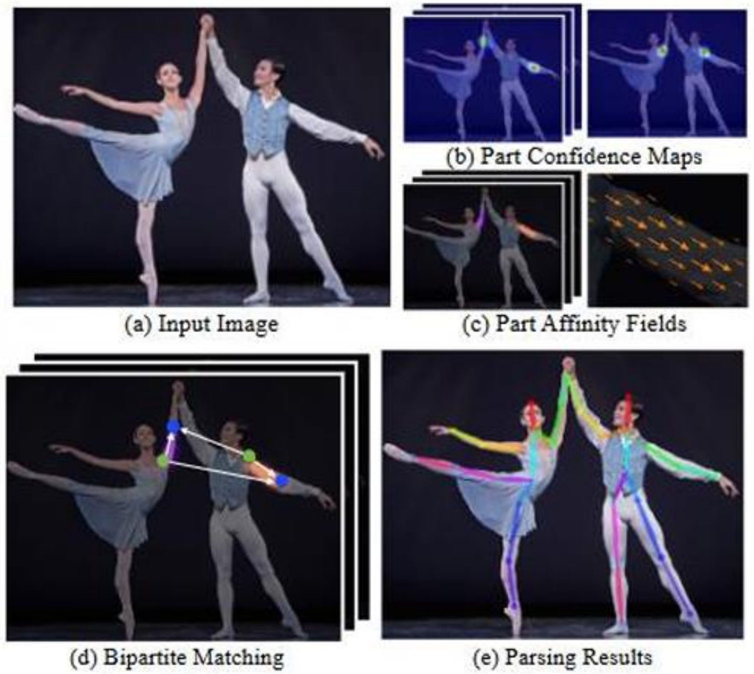


Pose Estimation - Open Pose

- **Open Pose** is a popular real-time pose estimation model.

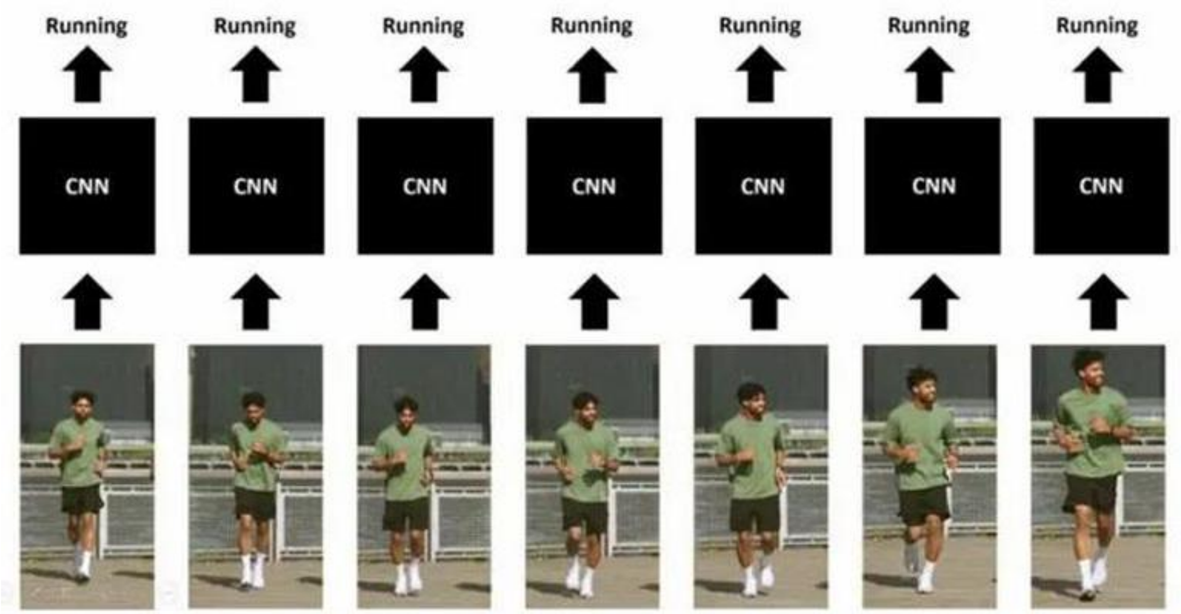
1. Identify *key points* (nose, elbows, knees, etc.) using **Part Affinity Fields**.
2. Repeat and refine multiple times using confidence maps.
3. Create a bipartite matching to connect the key points.





Temporal Modeling

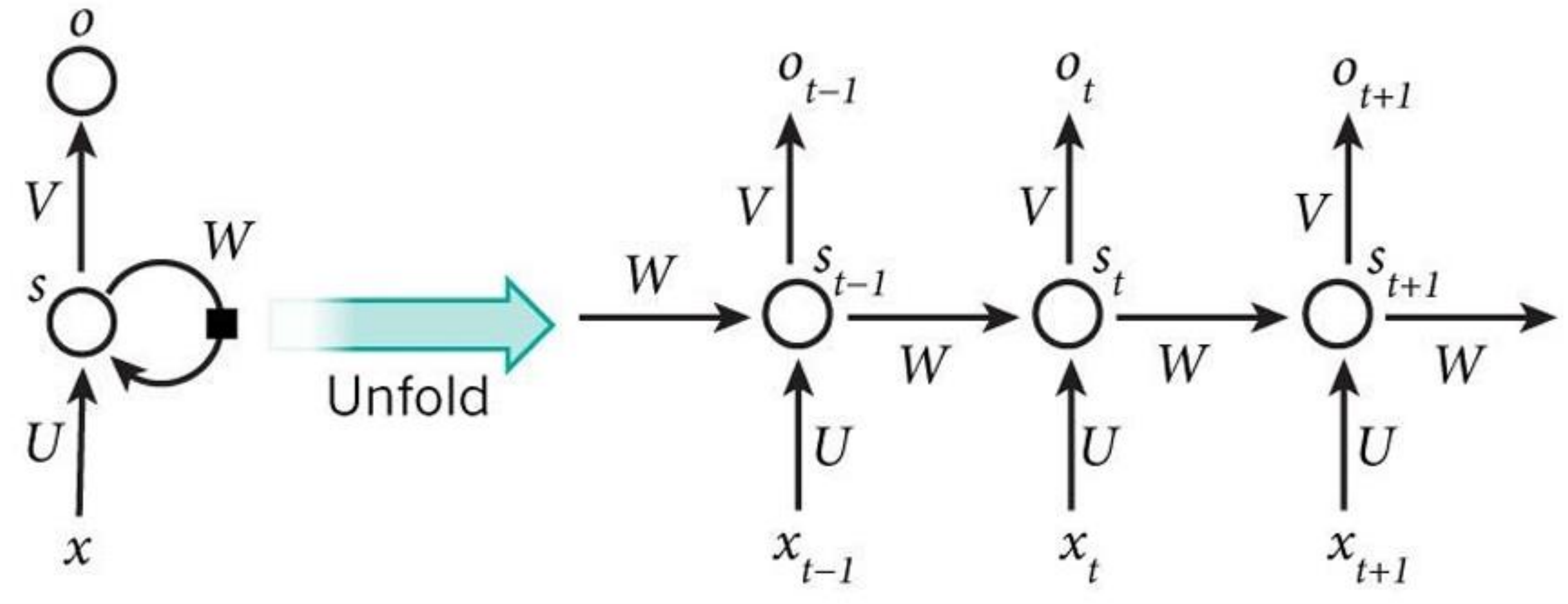
- Video classification can be done through frame-level image classification:



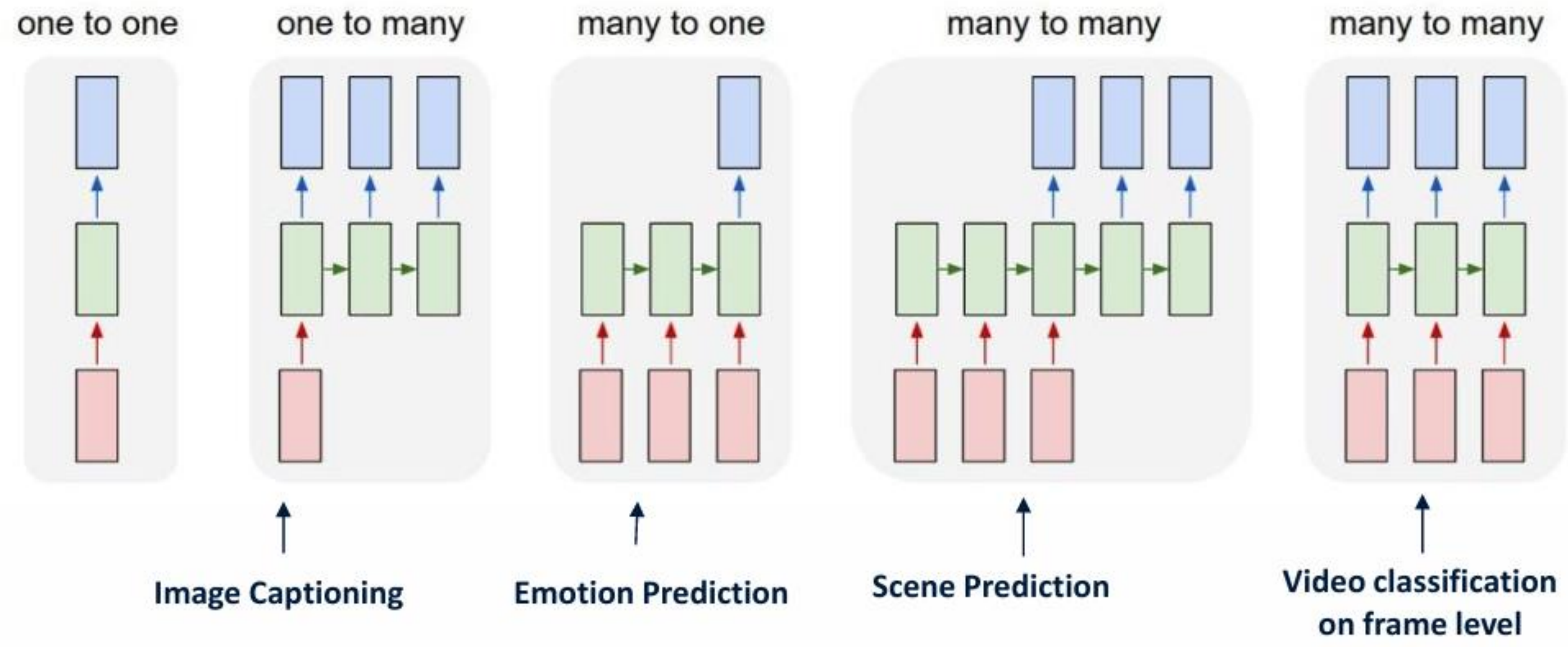
- But this fails to use the temporal information of a video.

Temporal Modeling - RNNs

- **Recurrent Neural Networks (RNN)** can process variable-length sequences.
- They are called *recurrent* since the same parameters are used at every step.

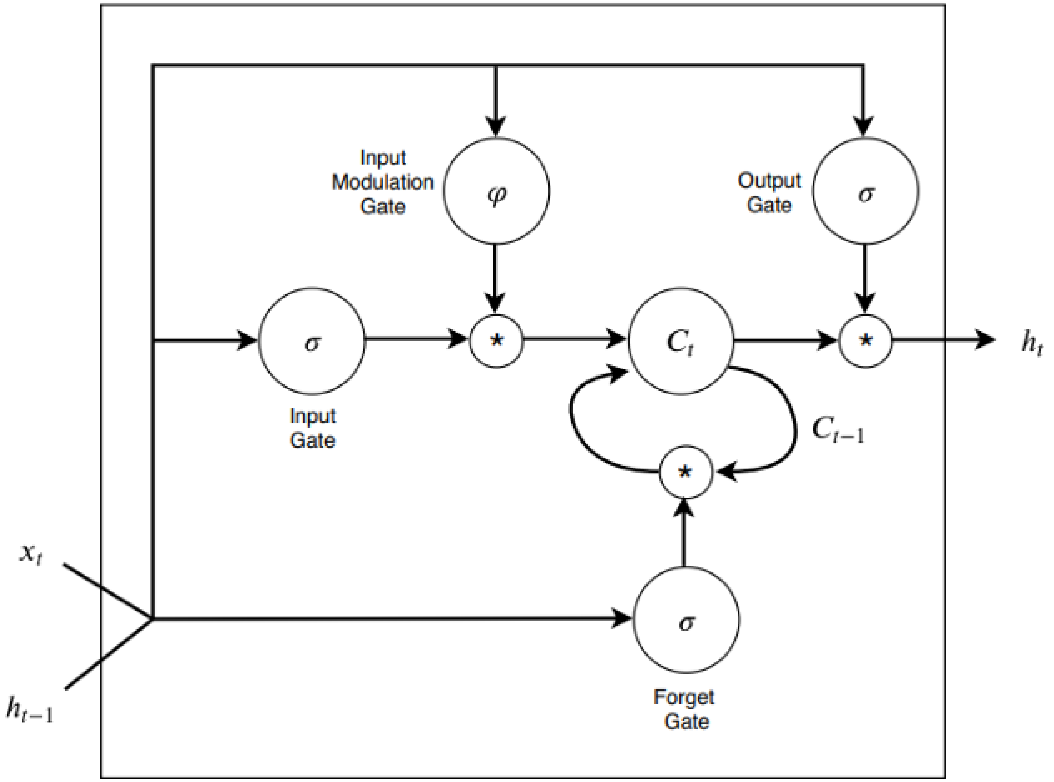


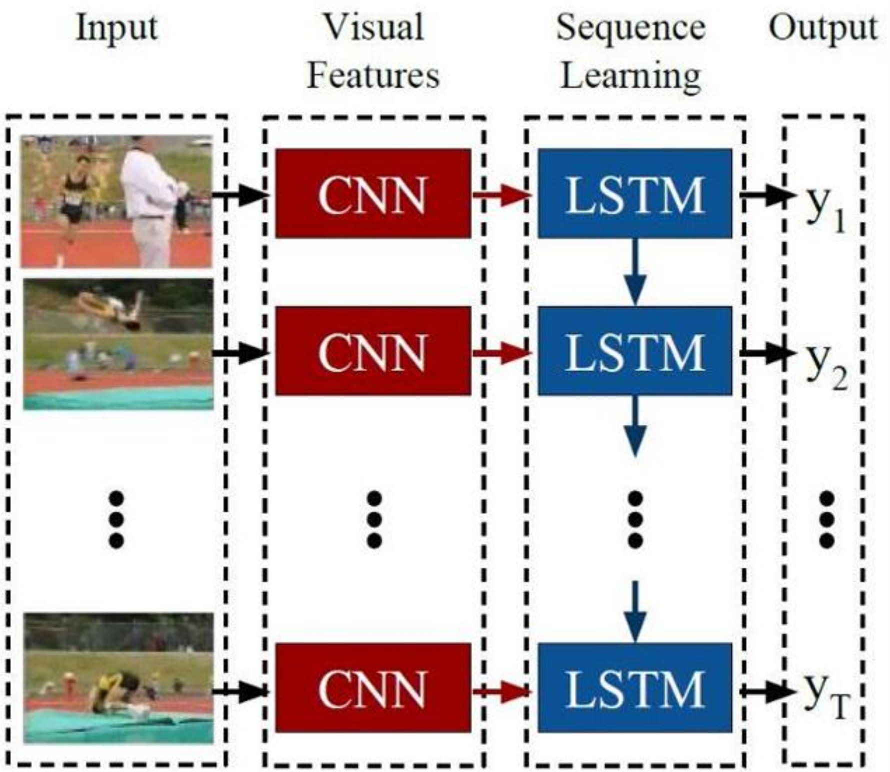
Temporal Modeling - RNNs



Temporal Modeling - LSTM Network

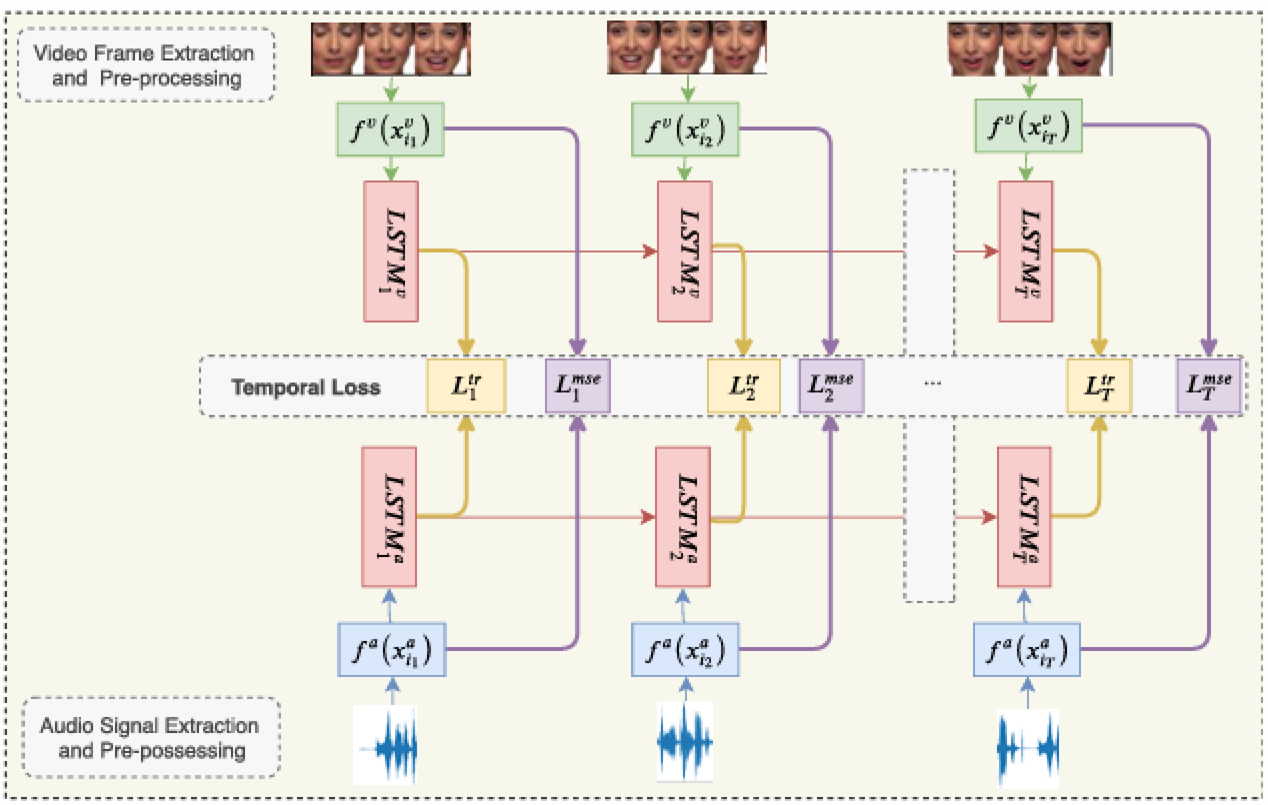
- **Long Short-Term Memory (LSTM)** networks are an improvement to RNNs.
- They overcome the *vanishing gradient* problem RNNs suffer from by learning to manage memory (*cell state*) through *input-*, *output-*, and *forget-gates*.

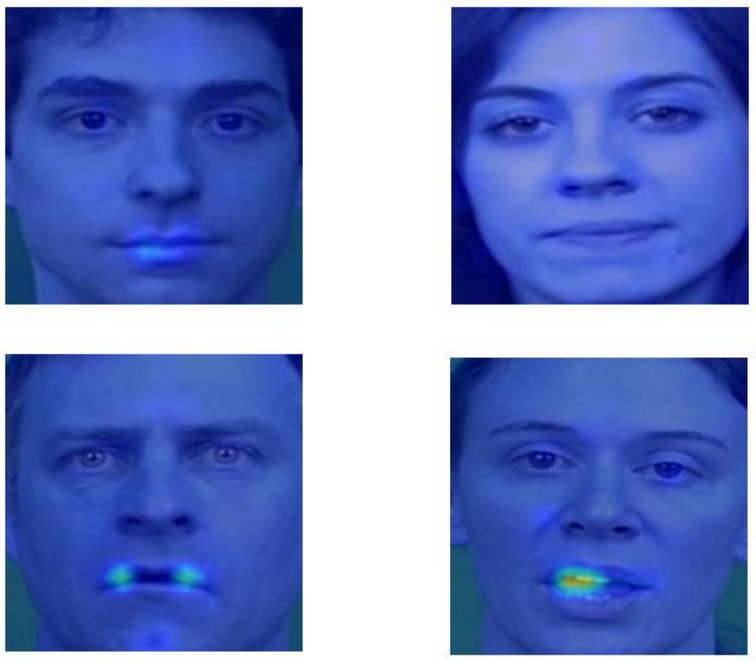




Temporal Modeling - LSTM Network

- LSTM Networks can be used to combine visual- and audio-processing:

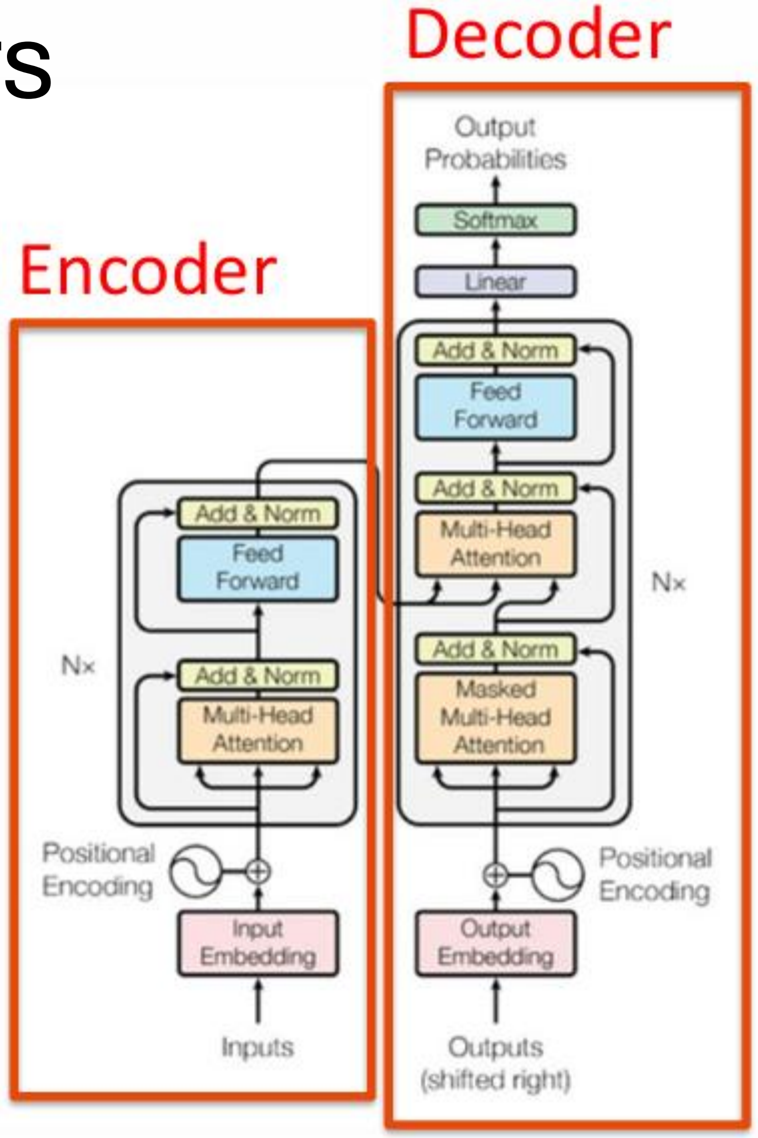




(Emotion Recognition)

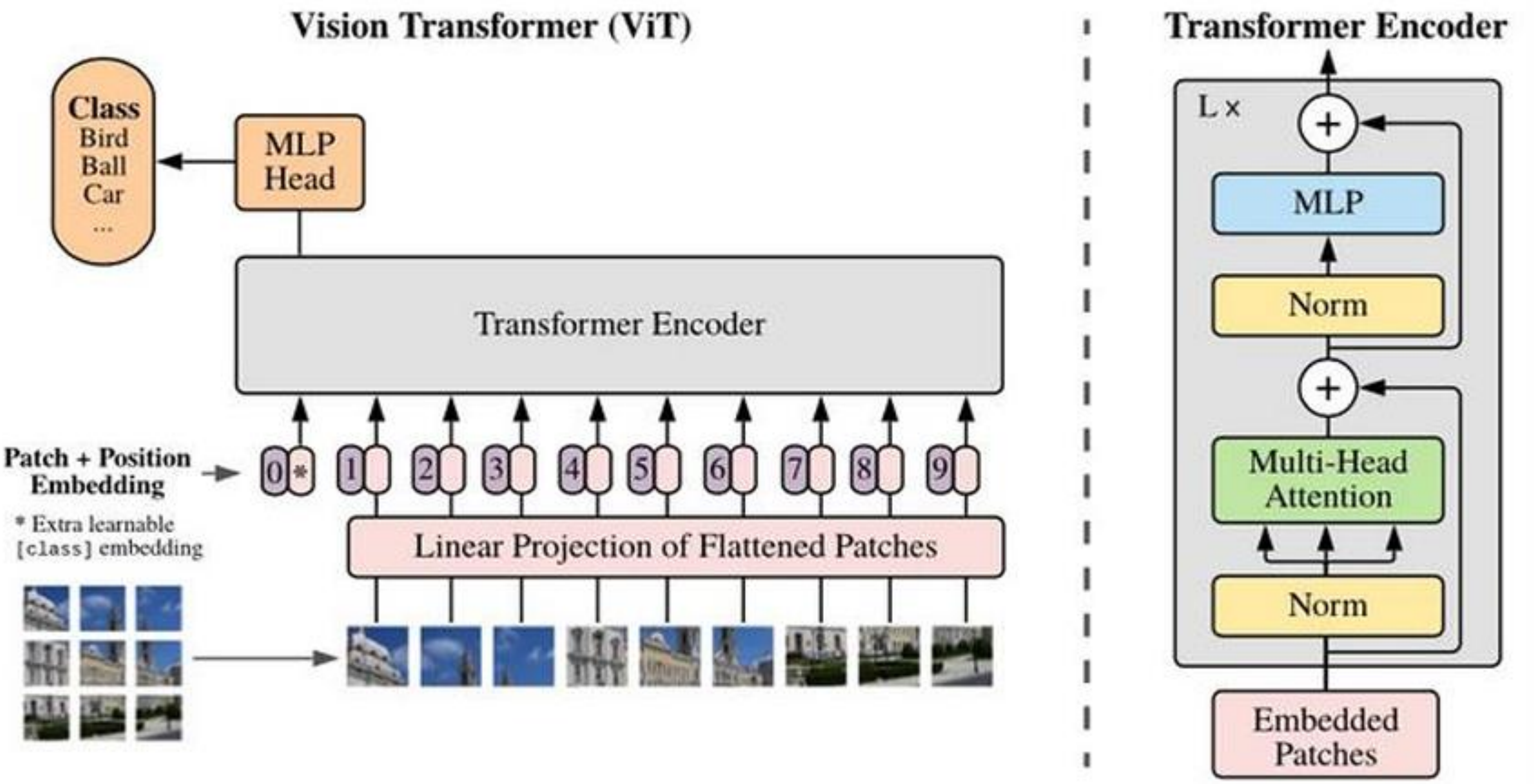
Temporal Modeling - Transformers

- Uses *attention* to process variable-length inputs.
- **Encoder**: learns a representation of the input.
- **Decoder**: decodes encoded representation and combines it with other input to predict the next token.



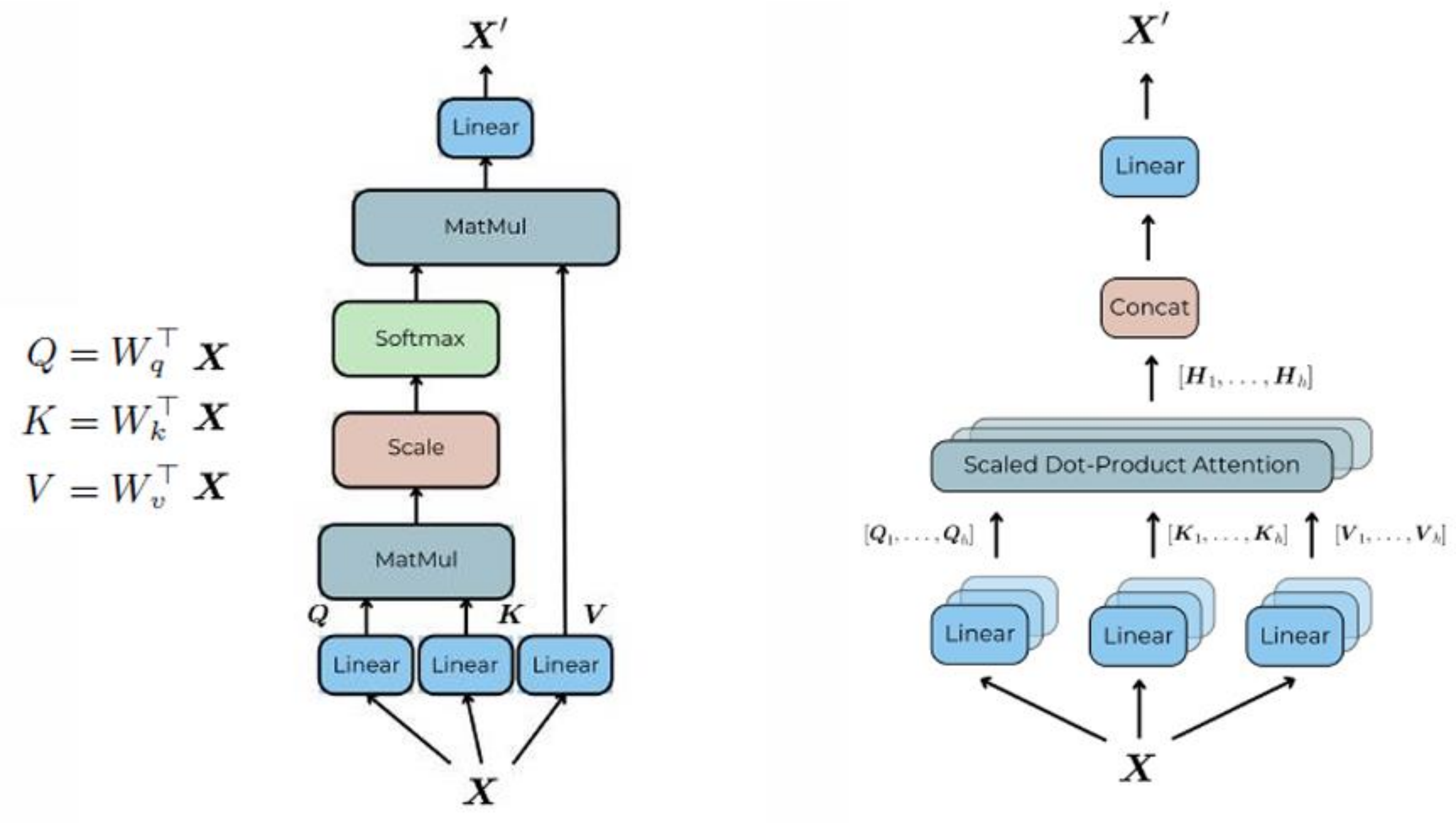
Vision Transformers

- An encoder-only **Vision Transformer** for classifying images:



Residual connections!

Vision Transformers - Attention



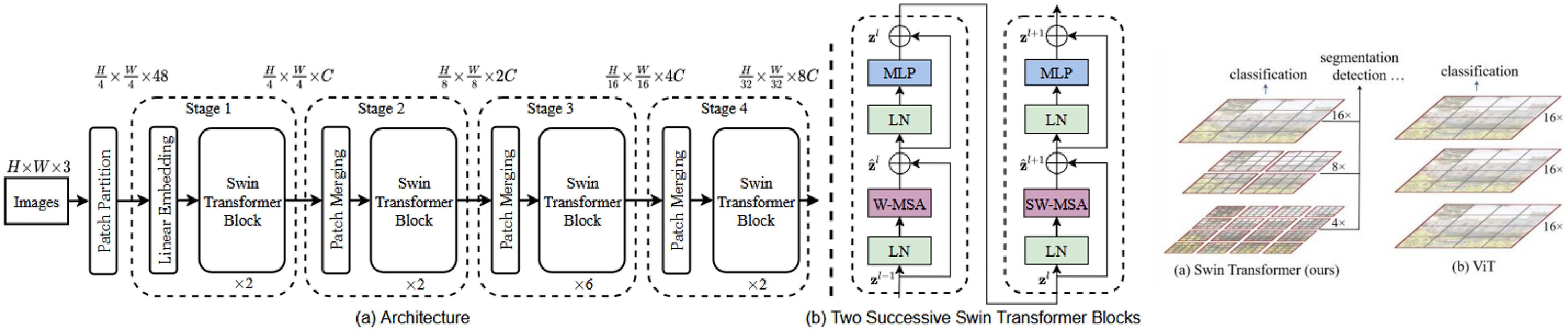
(a) Scaled dot-product attention

(b) Multi-Head Self Attention

- **Self-Attention**: use Q,K,V from the same sequence.
- **Cross-Attention**: uses Q from one sequence and K,V from another.
- **Causal-Attention**: ensure no information is given about future tokens.

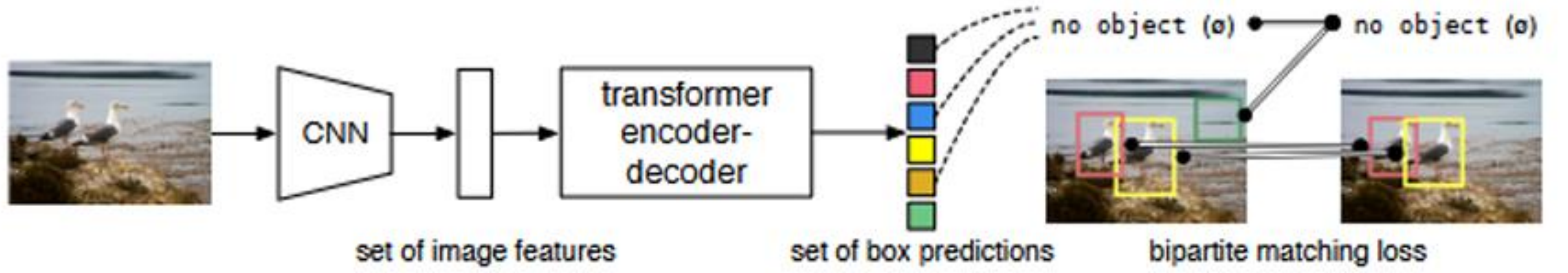
Vision Transformers - Swin Transformer

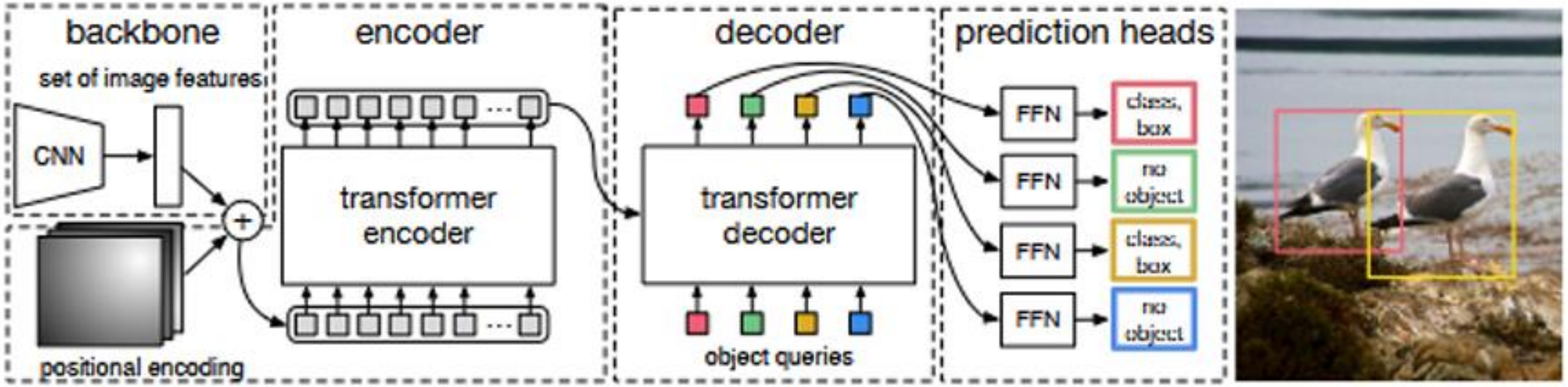
- The **Swin Transformer** is a general architecture that uses *shifted windows*:



Vision Transformers - DETR

- **End-to-End Object Detection with Transformers (DETR):**

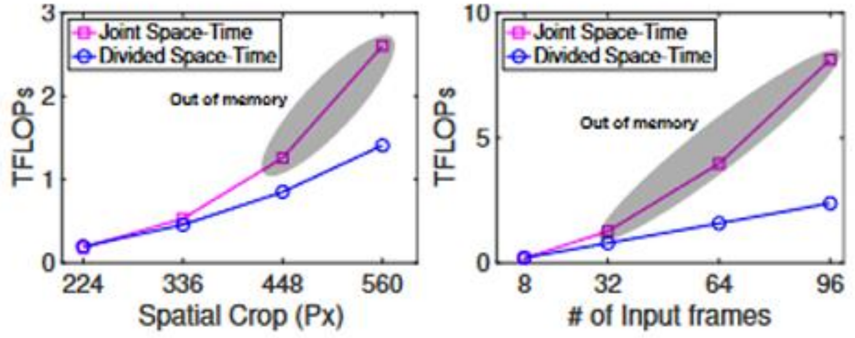


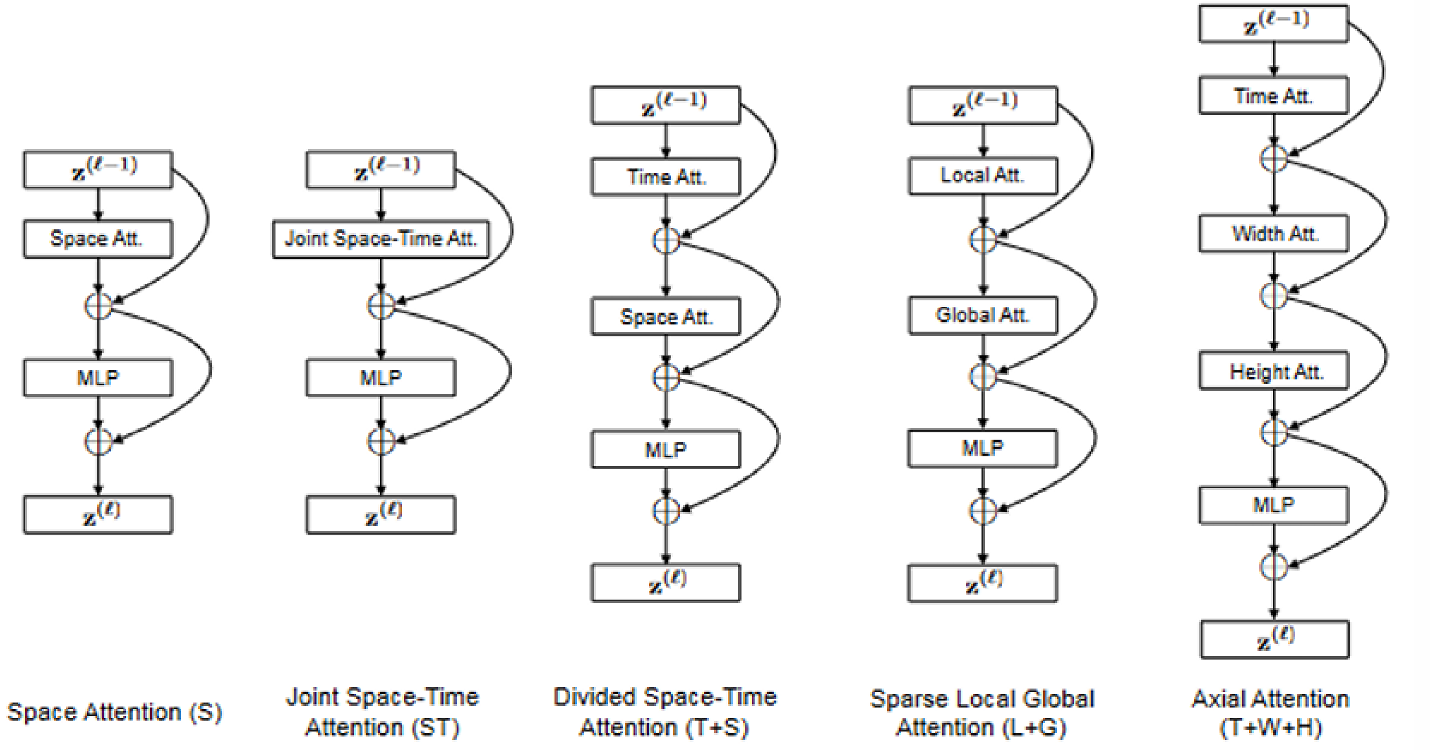


Video-Vision Transformers - TimeSformer

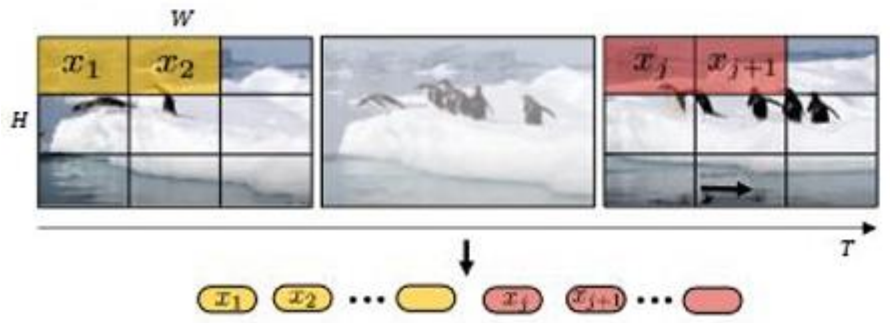
- When processing videos, **Video-Vision Transformers (ViVIT)** can make use of **Space-Time Self-Attention (TimeSformer)**.

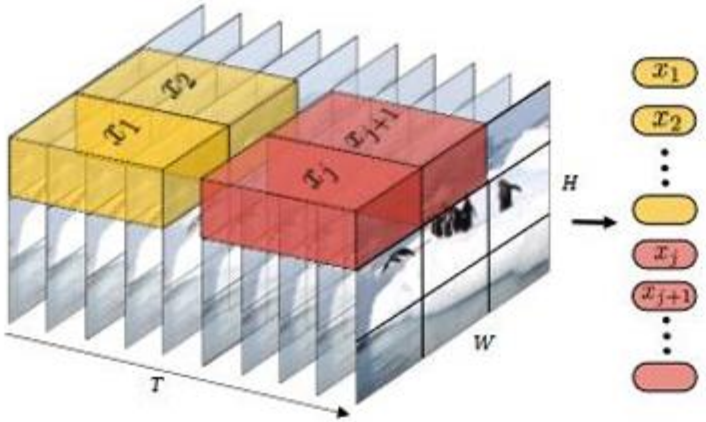
Attention	Params	K400	SSv2
Space	85.9M	76.9	36.6
Joint Space-Time	85.9M	77.4	58.5
Divided Space-Time	121.4M	**78.0**	**59.5**
Sparse Local Global	121.4M	75.9	56.3
Axial	156.8M	73.5	56.2

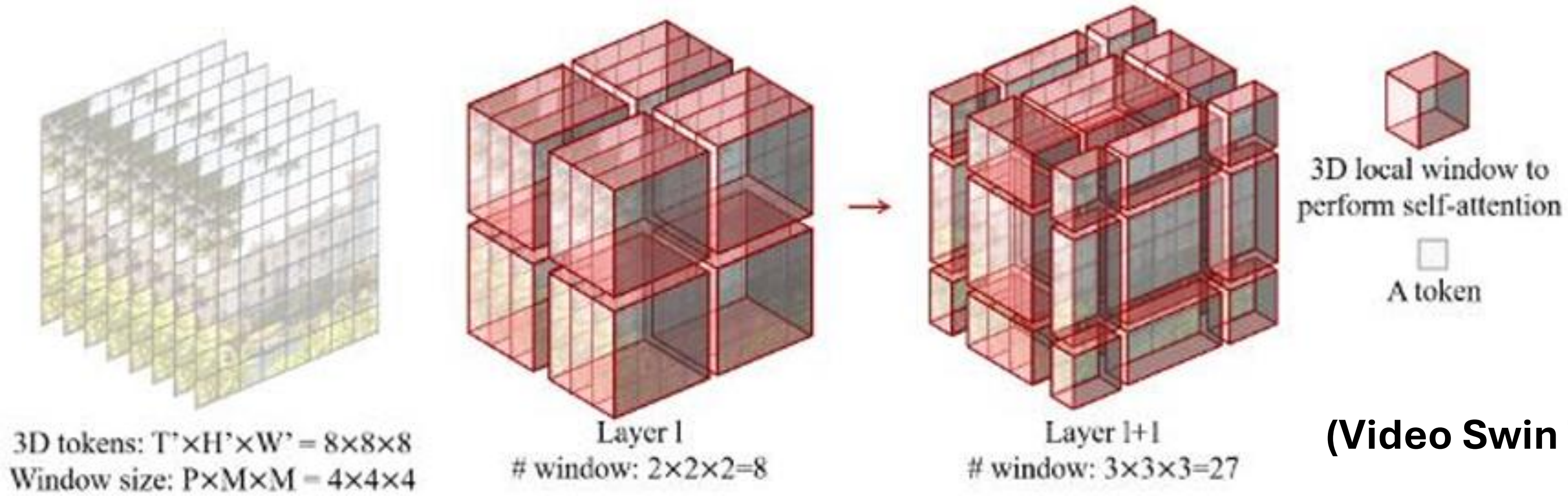




Video-Vision Transformers



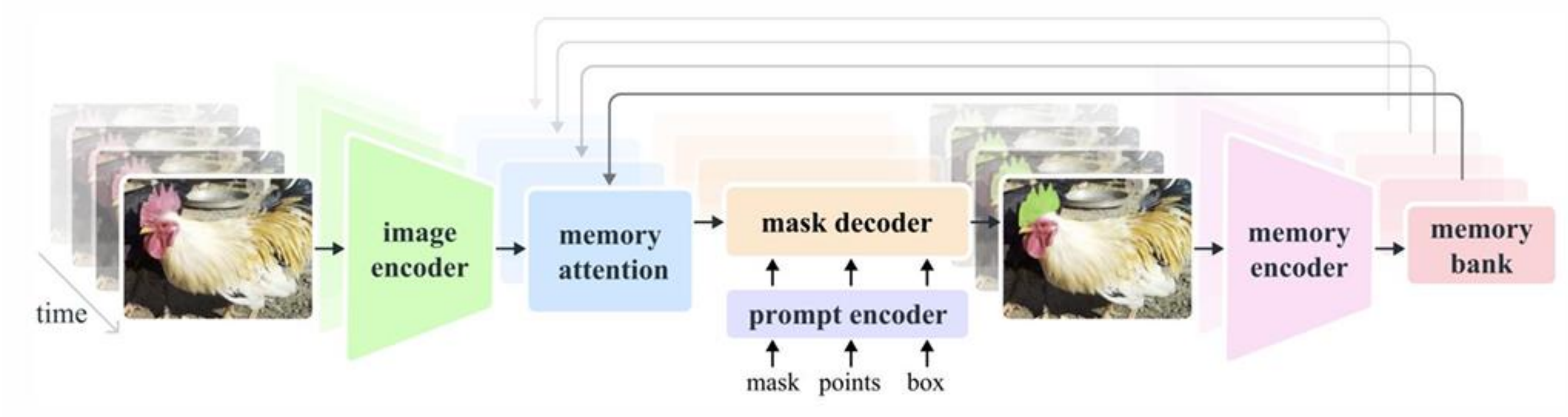




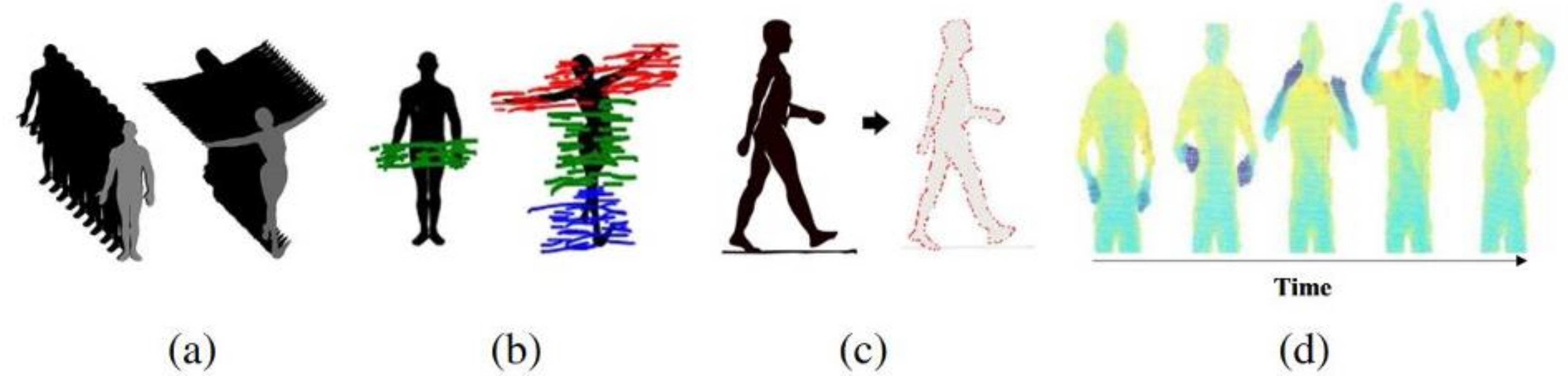
(Video Swin Transformer)

Video-Vision Transformers - SAM2

- **SAM2** can segment images and videos (in a frame-by-frame loop).



Activity Recognition

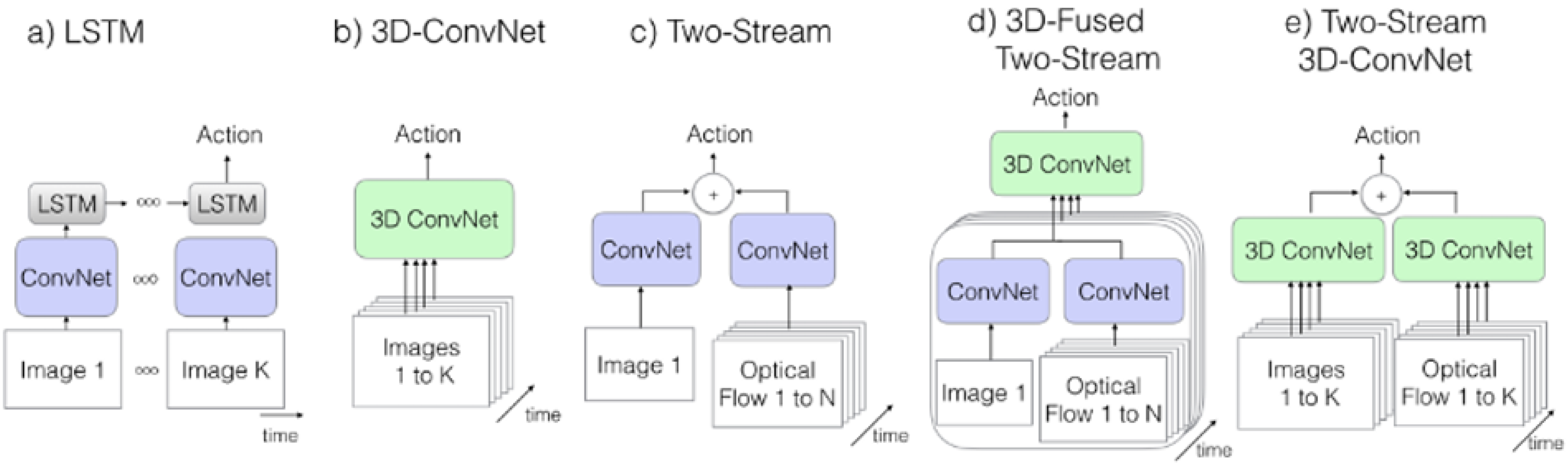


Examples of handcrafted feature extraction approaches: **a** Space-time volumes, **b** Space-time trajectories, **c** Shape-based methods: Contour features, **d** Motion-based methods [141]

Detection Complexity

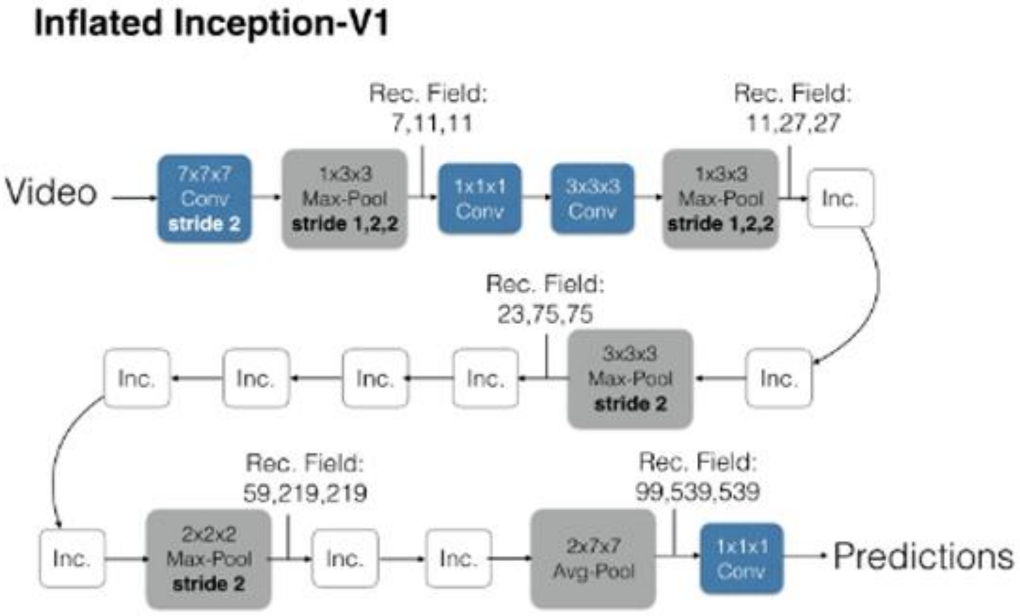
Activity Recognition - Baselines

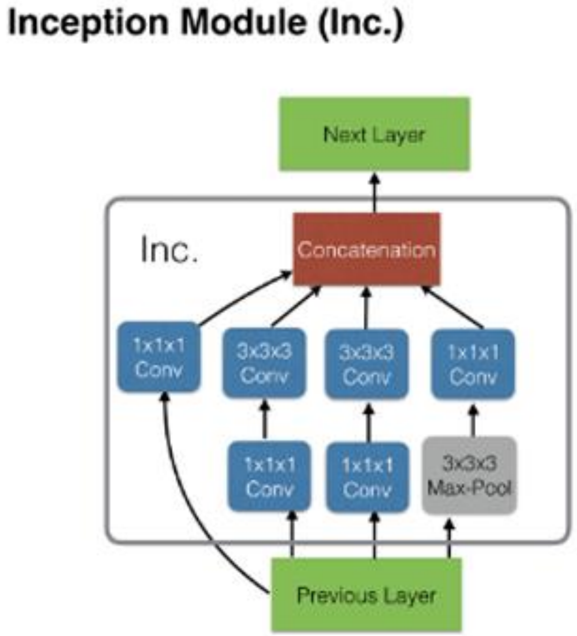
- Common baseline architectures for activity recognition:

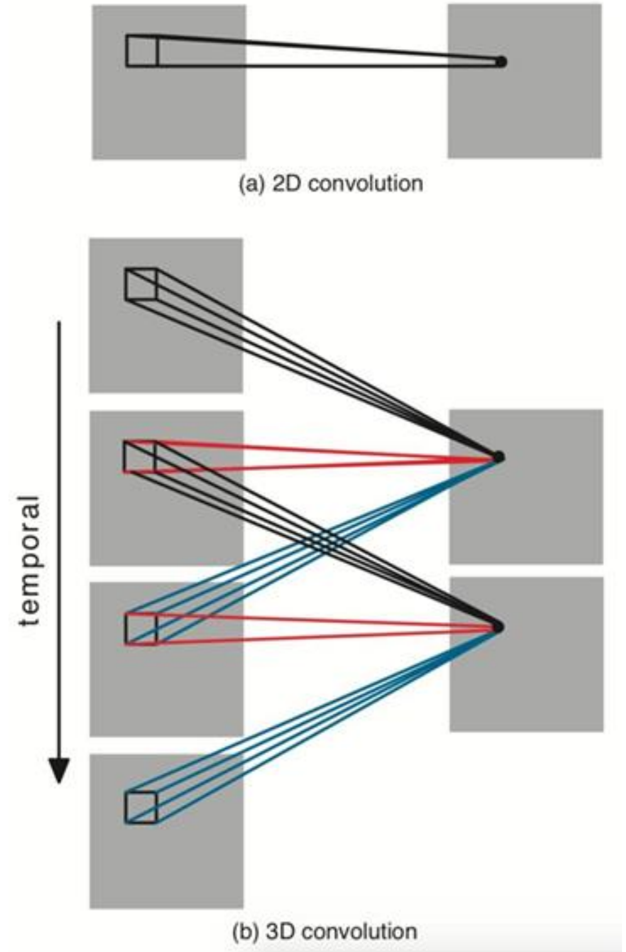


Activity Recognition - I3D

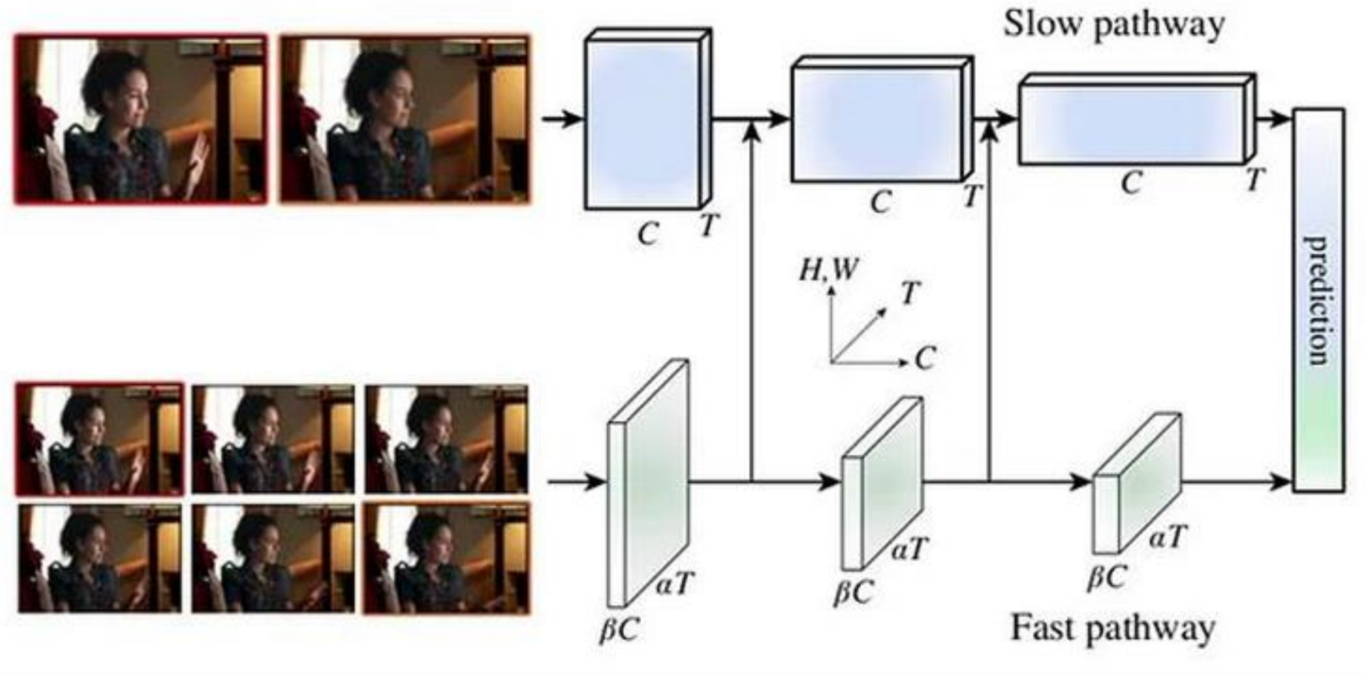
- **I3D** uses *3D filters* in combination with *inception modules*:







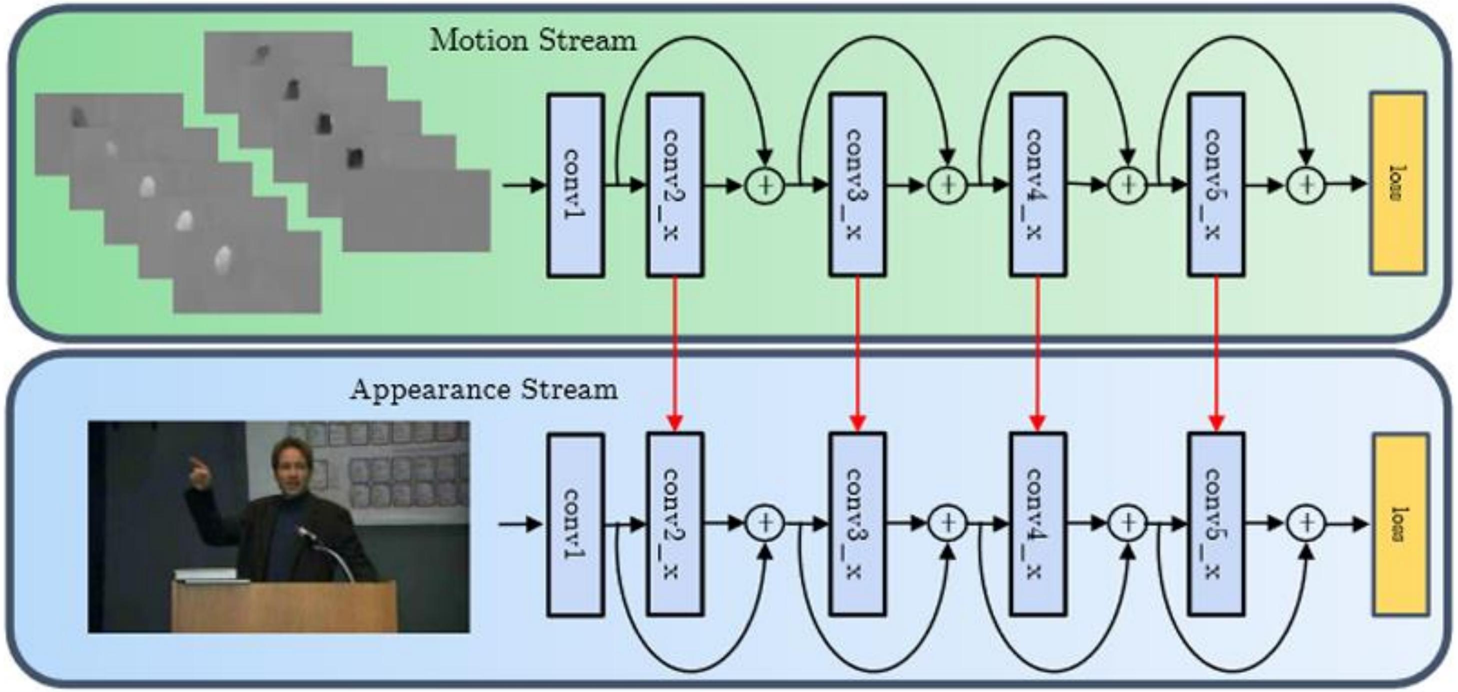
Activity Recognition - SlowFast



	Top-1 result	Top-5 result
Slow-Only	72.6%	90.3%
Fast-Only	51.7%	78.5%
Previous State-of-the-art	73.9%	90.9%
SlowFast	79.0%	93.6%

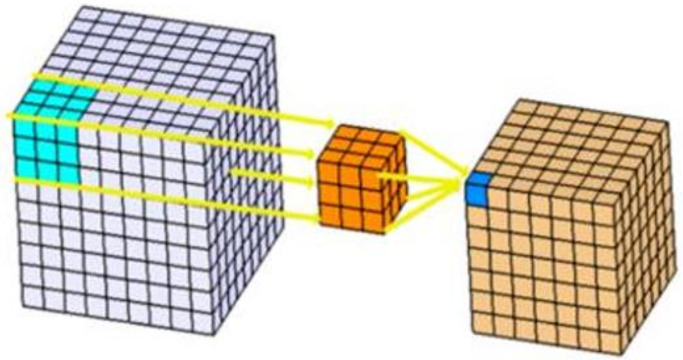
Activity Recognition - ST-ResNet

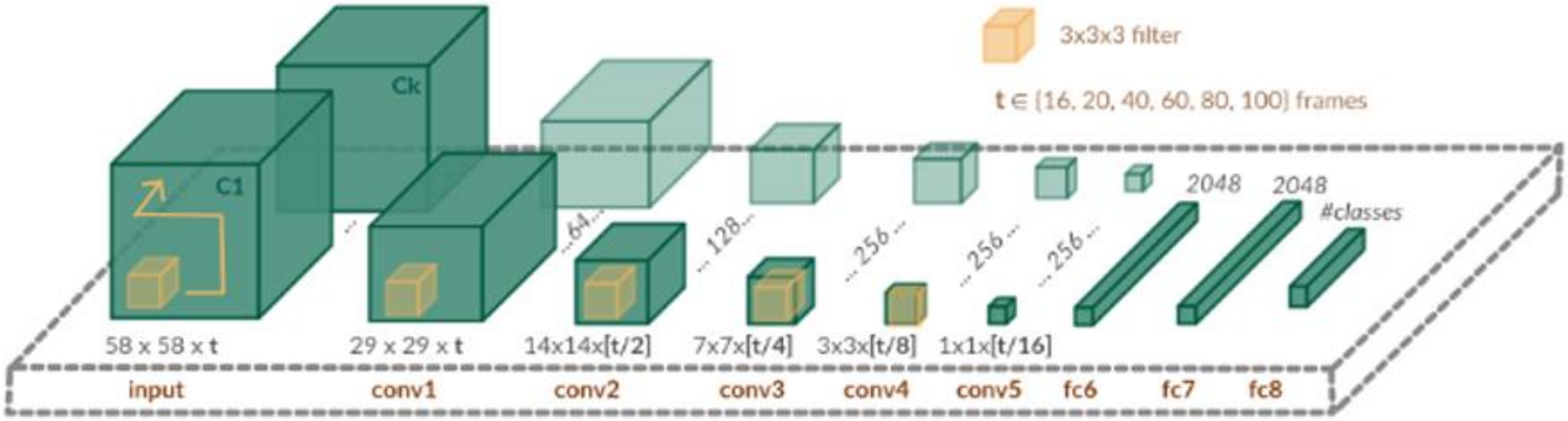
- Introduces *spatio-temporal residual connections* in a two-stream CNN:



Activity Recognition - 3D CNNs

- 3D CNNs work with 3D filters:





Temporal Grounding (Moment Retrieval)